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Abstract

The **Hospital Bed and Operating Room (OR) Scheduling System** is a software-based solution developed to efficiently manage hospital beds and operating rooms. In many hospitals, these resources are managed manually, which can result in scheduling conflicts, delays, inefficient resource utilization, and difficulties in handling emergency cases.

The proposed system provides a centralized platform where hospital staff can monitor bed availability, allocate beds to patients, and schedule surgeries based on operating-room availability, doctor availability, surgery duration, and patient priority. The system helps prevent double booking and improves coordination between different hospital departments.

By automating the scheduling process, the system aims to reduce patient waiting time, minimize scheduling errors, improve the utilization of hospital resources, and support better hospital management. Overall, the system provides an organized and efficient approach to hospital bed and operating-room scheduling.

Introduction

Hospitals require efficient management of patients, doctors, beds, and operating rooms to ensure timely treatment and effective use of resources. However, many hospitals still depend on manual records, spreadsheets, or separate systems for bed allocation and OR scheduling, which can cause **double booking, scheduling conflicts, surgery delays, improper bed allocation, and resource underutilization**. The proposed **Hospital Bed and Operating Room (OR) Scheduling System** provides a centralized digital platform to manage patients, beds, doctors, surgeries, operating rooms, and schedules. It helps administrators check resource availability and allocate them based on patient requirements and priority, including emergency cases. Overall, the system reduces manual effort, improves coordination, minimizes delays and conflicts, and ensures better utilization of hospital resources.

Literature Survey

Existing hospital management systems mainly focus on **bed management and Operating Room (OR) scheduling separately** to improve healthcare resource utilization. Bed management systems help identify available and occupied beds, while OR scheduling considers factors such as surgeon availability, operating-room capacity, equipment, surgery duration, and patient priority. Manual or disconnected systems can lead to delays, scheduling conflicts, and inaccurate resource information. Advanced optimization and AI-based approaches can improve scheduling efficiency but may require complex algorithms, large datasets, and higher implementation costs. Therefore, the proposed **Hospital Bed and Operating Room (OR) Scheduling System** integrates both functions into a single system, enabling better coordination, reducing conflicts and delays, improving resource utilization, and supporting faster decision-making by hospital administrators.

Problem Statement and Objectives

Problem Statement

Hospitals need to manage a limited number of beds and operating rooms while treating a large number of patients. In traditional systems, bed allocation and operating-room scheduling may be performed manually or through separate processes. This can result in **double booking, scheduling conflicts, delays in surgeries, improper bed allocation, and inefficient utilization of hospital resources**.

It can also be difficult for hospital administrators to quickly identify available beds and operating rooms, especially during emergency situations. Therefore, there is a need for an integrated scheduling system that can provide accurate availability information and efficiently allocate hospital resources according to patient needs and priority.

The proposed **Hospital Bed and Operating Room (OR) Scheduling System** aims to automate and simplify the scheduling process by providing a centralized platform for managing beds, operating rooms, patients, doctors, and surgery schedules.

Objectives:

The main objectives of the proposed system are:

1. To develop a centralized system for managing hospital beds and operating rooms.
2. To provide updated information about available and occupied beds.
3. To allocate beds to patients based on availability and requirements.
4. To schedule surgeries based on operating-room and doctor availability.
5. To prevent double booking and scheduling conflicts.
6. To give appropriate priority to emergency cases.
7. To reduce patient waiting time and scheduling delays.
8. To improve the utilization of hospital beds and operating rooms.
9. To reduce manual work and minimize scheduling errors.
10. To provide an easy-to-use platform for hospital administrators and staff.

Results

Results

The proposed **Hospital Bed and Operating Room (OR) Scheduling System** successfully provides a centralized platform for managing hospital beds and operating-room schedules. The system allows hospital administrators to monitor resource availability and assign beds and operating rooms according to patient requirements and scheduling priorities.

The system is expected to produce the following results:

- **Efficient Bed Management:** Hospital staff can easily view available, occupied, and reserved beds and allocate them to patients accordingly.
- **Effective OR Scheduling:** Operating rooms can be scheduled based on surgery requirements, doctor availability, and available time slots.
- **Reduced Scheduling Conflicts:** The system checks existing schedules to prevent double booking of beds, operating rooms, and doctors.
- **Improved Resource Utilization:** Better scheduling helps reduce unused beds and operating-room time.
- **Emergency Case Management:** Emergency surgeries and patients can be given higher priority when resources become available.
- **Reduced Manual Work:** Automated scheduling reduces dependence on paper records and manual calculations.
- **Faster Decision-Making:** Administrators can access updated resource information from a centralized dashboard.
- **Better Coordination:** The system improves coordination between hospital administrators, doctors, nurses, and other departments.

Overall, the system provides a more **organized, efficient, and reliable approach to hospital resource scheduling**, helping reduce delays and improve the utilization of critical hospital resources.

Hardware and Software requirements

Hardware Requirements::

- **Processor:** Intel Core i3 or equivalent and above
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 10 GB of free storage
- **Display:** Standard monitor or laptop display
- **Input Devices:** Keyboard and mouse
- **Network:** Internet/LAN connection for accessing the system and database
- **Server:** Local or cloud server for storing and managing application data

Software Requirements:

- **Operating System:** Windows 10/11 or Linux
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python / Java / Node.js
- **Database:** MySQL / MongoDB
- **Development Environment:** Visual Studio Code
- **Web Browser:** Google Chrome / Microsoft Edge / Mozilla Firefox
- **Version Control:** Git and GitHub
- **Server Environment:** XAMPP / Node.js / Python server, depending on the selected backend

Conclusion

Conclusion

The **Hospital Bed and Operating Room (OR) Scheduling System** provides an efficient and organized solution for managing hospital beds and operating-room schedules. The system reduces the difficulties associated with manual scheduling by providing a centralized platform to monitor resource availability, allocate beds, and schedule surgeries.

By considering factors such as **bed availability, operating-room availability, doctor schedules, surgery duration, and patient priority**, the system helps reduce scheduling conflicts and improve the utilization of hospital resources. It also supports faster decision-making and better coordination among hospital staff.

The proposed system can help minimize waiting time, reduce manual errors, and improve the overall efficiency of hospital operations. Therefore, the system provides a practical approach to managing critical hospital resources and can contribute to better and more timely patient care.

Future Scope

Future Scope:

The system can be further enhanced with advanced features to improve hospital resource management.

- **AI-based scheduling** for generating optimized bed and OR schedules.
- **Real-time notifications** for doctors, nurses, and hospital staff.
- **Mobile application** for easy access to schedules and bed availability.
- **Emergency priority management** for automatically handling urgent cases.
- **IoT integration** for real-time monitoring of bed occupancy.
- **Cloud integration** for secure access and centralized data management.

These improvements can make the system more **intelligent, flexible, and efficient** in managing hospital resources.

Thank you